

Emily Alger

The Big Data Institute, University of Oxford, Old Road Campus, Oxford, OX3 7LF

[Website](#)

Email: emily.alger@ndm.ox.ac.uk

[Google Scholar](#)

I develop Bayesian adaptive clinical trial methodology, with a particular focus on dose-finding and patient-centred trial design.

Professional position

November 2025 – Present: University of Oxford, Postdoctoral Research Assistant

Oxford-GSK Centre for Biostatistics and AI in Medicine (CeBAM).

- Developing novel Bayesian adaptive methodology to improve sequential data collection and decision-making within clinical trials.

Education

October 2022 – October 2025: Institute of Cancer Research, PhD in Biostatistics

- Supervised by Professor Christina Yap, degree awarded 27th April 2026.
- Thesis: *Integrating Patient-reported Outcomes (PROs) in Early Phase Dose-finding Oncology Trials*

October 2017 – June 2022: University of Warwick, Mathematics and Statistics (BSc MMathStat)

- First Class - Bachelor of Science, Master of Mathematics and Statistics (with Honours).
- Relevant final year modules: Monte Carlo Methods (93), Statistical Learning and Big Data (82), Bayesian Statistics and Decision Theory (82), Bayesian Forecasting and Intervention (76).

Selected First-Author Publications:

- Alger, E., Aiyegbusi, O. L., Dueck, A. C., Minchom, A., Pe, M., Peipert, J. D., ... & Yap, C. (2026). [International consensus-driven recommendations for patient-reported outcome research objectives in early phase dose-finding oncology trials: OPTIMISE-ROR](#). *Journal of Clinical Oncology*, **44**(8), 709-719.
- Alger, E., Regnault, A., Dueck, A. C., Pe, M., Grayling, M. J., Calvert, M. J., ... & Yap, C. (2026). [A practical toolkit with recommendations for analysing and visualising patient-reported outcomes in early phase dose-finding oncology trials \(OPTIMISE-AR\)](#). *The Lancet Oncology*, **27**(4), e218-e230.
- Alger, E., Lee, S. M., Cheung, Y. K. K., & Yap, C. (2026). [Calibration of Priors for Bayesian Model-Based Dose-Finding Trial Designs With Joint Outcomes](#). *Statistics in Medicine*, **45**(23-24), e70746.
- Alger, E., Mandrekar, S. J., Yin, J., & Yap, C. (2026). [PRO-ADD: Patient-empowered dose-finding trials integrating safety, preliminary efficacy and patient-reported outcomes for optimal dose selection](#). *Statistical Methods in Medical Research*, 09622802261435969.
- Alger, E., Lin, R., Lee, J. J., Yuan, Y., & Yap, C. (2026). [Generalized Pairwise Comparisons in Dose Optimization Oncology Trials: Beyond Safety to Multi-outcome Dose Selection](#). *Clinical Cancer Research*, **32**(13), 2787-2796.

Grants and Awards

- **May 2025: American Statistical Association Biopharmaceutical Section Scholarship Award**, recognising notable research and academic achievement within biopharmaceutical statistics.
- **May 2025: ESMO Merit Travel Grant**, Travel bursary to present at ESMO TAT Asia Congress 2025.
- **February 2025: Institute of Cancer Research Undergraduate Summer Scholarship Scheme**, secured funding to supervise an undergraduate summer student in summer 2025.
- **October 2024: TMRP Student Network Funding**, Travel grant to visit MD Anderson Cancer Centre in October 2024 to research adaptive borrowing techniques for Patient-Reported Outcomes.
- **May 2024: Student award, Advanced Statistical Designs to Empower Biomarker-driven Clinical Trials** workshop, Travel bursary to present at the University of Bath.

- **May 2024: Training Support Fund, The Alan Turing Institute**, Travel bursary to support my oral presentation at the Society for Clinical Trials annual general meeting in May 2024.
- **October 2023: Enrichment Scheme award, The Alan Turing Institute, London, UK**
- **March 2023: ESMO Merit Travel Grant**, Travel bursary to present at ESMO TAT 2023 in March 2023.

Selected Conference Presentations

- **June 2026 – Invited Talk, PSI Conference**, Belfast, UK: *Inclusion of PROs in Dose-Finding Oncology Trials.*
- **July 2025 – Talk, ESMO TAT Asia Congress**, Hong Kong SAR, China: *International consensus-driven recommendations for patient-reported outcome research objectives in early phase dose-finding oncology clinical trials: OPTIMISE-ROR*
- **June 2025 – Poster, BayesComp**, Singapore: *Calibration of Priors for Bayesian Model-Based Dose-Finding Trial Designs With Joint Outcomes.*
- **May 2025 – Talk, 46th Annual Meeting of the Society for Clinical Trials**, Vancouver, Canada: *Calibration of dose-agnostic priors for Bayesian dose-finding trial designs with joint outcomes.*
- **July 2024 – Talk, 45th Annual Conference of the International Society for Clinical Biostatistics**, Thessaloniki, Greece: *Patient-Centred Dose-Finding Trials using Safety, Efficacy and Patient-Reported Outcomes.*

Research visits

October 2024: MD Anderson Cancer Centre, Houston, US.

- 10-day research visit to develop novel Phase I/II trial design to incorporate Patient-Reported Outcomes with Professor Jack Lee, Professor Ying Yuan and Dr Ruitao Lin.

May 2023: Columbia University, Mailman School of Public Health, New York, US.

- 3-week research visit to develop novel dose-finding trial designs which incorporate Patient-Reported Outcomes with Dr Shing Lee and Professor Ying Kuen Cheung.
- Associated paper published in [ESMO Open](#).

Academic Service

- **Section Committee Member**, NIHR Statistics Group, Early Phase Clinical Trials.
- **Peer reviewer** for *Annals of Applied Statistics*, *Statistics in Medicine*, *Pharmaceutical Statistics*, *Nature Communications*, and *ESMO Open*.
- **Gold Reviewer** for The International Conference on Machine Learning (ICML 2026).
- **On-site organiser** of the Trials Methodology and Research Partnership's Adaptive Design Working group in-person meeting, 16th-17th April 2024, hosted at the Institute of Cancer Research.

Experience

October 2023 – July 2024: Enrichment Student, The Alan Turing Institute

- Established Bayesian Statistical Methods reading group: Generalised Bayes, Martingale posteriors.
- Collaborated with Academic services as an Enrichment student blogger.
- Successfully obtained funding to run student events to promote engagement and cohort building among the Turing community as part of my Community Champion role.

July – September 2022: Research Assistant, MRC Biostatistics Unit, University of Cambridge

- Investigated Restricted Mean Survival Time endpoint within clinical trial design.
- Simulated fixed sample and group sequential clinical trials in R using cluster computing techniques.
- Associated paper published in *Trials*, 2026: [Estimation of combined treatment effects by restricted mean survival time](#).

July – August 2021: Summer vacation student within Early Phase and Adaptive Trials, The Institute of Cancer Research

- Associated paper published in *Contemporary Clinical trials*, 2023: [Reporting quality of CONSORT flow diagrams in published early phase dose-finding clinical trial reports: Improvement is needed](#).